Structural Non-Isolability in Knowledge-Based Robot Fault Diagnosis

Chenxuan Zhou * **Jie Liang** *

* *Faculty of Engineering and Information Technology, University of Technology Sydney, Ultimo, NSW 2007, Australia (e-mail: chenxuan.zhou@student.uts.edu.au, jie.liang@uts.edu.au).*

Abstract: A robot diagnosis layer scores 14 failure modes of our FMEA knowledge base against per-signal deviations. On the voraus-AD benchmark, two or three modes tie at the top in 514 of 755 anomalous cycles, always within one of four groups that declare the same joints and signal families. Top-1 accuracy ranges from 0.119 to 0.200 over tie-break orders; a prevalence prior reaches 0.187. We report the non-isolable group and search for statistics that separate it. For the load-torque group they raise held-out within-group accuracy from 0.562 to 0.840. For the collision group, random-split validation adopts statistics that leave-one-setting-out validation rejects.

Keywords: Fault isolation and root cause analysis, Structural analysis methods, Diagnosability analysis, Knowledge-based diagnosis, Anomaly detection, Robotics.

1. INTRODUCTION

An anomaly detector on robot telemetry can tell that a cycle is abnormal, but not why. To name likely causes, a line-specific knowledge base is often built from a failure mode and effects analysis (FMEA). Each failure mode declares the joints it affects and the signal families in which it appears, a retrieval function scores the modes against the per-signal deviations of the cycle, and the maintenance technician receives a ranked list of candidate causes.

We built such a layer for the public voraus-AD benchmark (Brockmann et al., 2024) and asked which failure modes its list can isolate. The declarations of the knowledge base form a fault signature matrix in the sense of Blanke et al. (2016). In this matrix, nine of the 14 failure modes fall into four groups whose members declare the same joints and signal families. In three groups the columns are identical; in the fourth, one column contains the other two. The scoring function is continuous, yet in 514 of 755 anomalies two or three modes of one group receive the same top score. Which of them is reported then depends only on the order in which the modes are listed (Sec. 5.2).

These ties affect how the layer is evaluated. Its top-1 accuracy depends on the tie-break order and ranges from 0.119 to 0.200 over all orders. The reference for judging accuracy also matters. During development we compared top-1 accuracy with uniform chance. When fault frequencies are skewed, a more useful reference is a prevalence prior, which ranks the modes and groups by their frequency in the training data and gives the same ranking for every cycle. On voraus-AD this prior is better than the retrieval on top-3, top-5, mean reciprocal rank (MRR), and group accuracy under every tie-break order (Sec. 5).

The ties also lead us to ask whether statistics of the signals that are already recorded can separate the members of a group. For each non-isolable group we search for a set of such statistics, which we call a *repair*. For the load-torque group a repair raises held-out accuracy within the group from 0.562 to 0.840. For the collision group the answer depends on how the search is validated. The `setting` field of voraus-AD identifies a physical fault instantiation, for example one added weight or one placement of a collision object, and each instantiation is repeated for 4 to 20 cycles. A random split over cycles puts cycles of the same instantiation into both training and test data. Under such a split the search adopts a collision repair that is rejected when whole instantiations are held out (Sec. 7). Leakage of this kind is known in machine-learning-based science (Kapoor and Narayanan, 2023) and in fault-diagnosis benchmarks (Hendriks et al., 2022); here it affects the inner validation that selects the repair. Because a repair is learned from labelled cycles, we also compare retrieval followed by the repair with a flat classifier trained on the same features and labels (Sec. 8).

2. FAULT SIGNATURES AND ISOLABILITY

Structural diagnosability analysis uses the structure of the model equations to determine, before any data are seen, which faults a diagnosis system can at best separate (Blanke et al., 2016; Krysander et al., 2008). Its standard object is the fault signature matrix, whose rows are tests or residuals and whose columns are fault modes. Under the exoneration assumption, two modes are non-isolable exactly when their columns coincide. Without it, isolability is directional: a mode whose column is contained in another mode's column cannot be isolated from that mode (Krysander and Frisk, 2008; Cordier et al., 2004). The analysis has been applied to public process benchmarks (Düstegör et al., 2006), extended to graded distin-

guishability under noise (Jung et al., 2015), and related to spectrum-based fault localisation in software (Travé-Massuyès and Wotawa, 2024).

When the signatures do not separate the faults, signature matching in fault detection and isolation (FDI) and consistency-based diagnosis in the artificial-intelligence (DX) tradition (Reiter, 1987; de Kleer and Williams, 1987) both return a set of candidates. For a complete set of tests without exoneration, the two sets coincide (Cordier et al., 2004; Travé-Massuyès, 2014). A ranked list that breaks ties names one member of this set as the answer and ranks the others below it. We therefore report the non-isolable group of the top-ranked mode, which contains every mode tied with it.

The usual remedy for a non-isolable group is sensor placement, which finds an additional measurement that makes the group isolable at the lowest cost (Krysander and Frisk, 2008; Frisk et al., 2009). Our case has no analytical model of the cell, and the rows of the signature matrix are declarations in an FMEA knowledge base (affected joints, signal families, cycle phase) rather than residuals of a model. Without a model, a candidate test can be judged only on data, and data exist only for the signals that were recorded. We therefore search among statistics of the recorded signals that the retrieval does not use. This is test selection with data-driven evaluation, close to the hybrid scheme of Jung et al. (2018). A statistic that needs a new measurement corresponds to sensor placement; the others use only signals the cell already records. The evidence for a statistic is its accuracy on held-out fault instantiations.

3. SYSTEM AND DATA

Benchmark. The voraus-AD dataset (Brockmann et al., 2024) contains machine data from a six-axis robot performing a pick-and-place task, with 12 categories of physically induced anomalies. We use the 100 Hz release, which has 2122 cycles: 755 anomalous and 1367 normal. The official training data are 948 normal cycles, and the remaining 419 normal cycles are held out. The anomalous cycles fall under 72 `setting` values. Each value identifies one physical fault instantiation, repeated for 4 to 20 cycles (median 10). Three categories have a single instantiation each.

Evidence record. We evaluate diagnosis on all 755 labelled anomalous cycles, which amounts to assuming perfect detection; in deployment an anomaly detector such as that of Brockmann et al. (2024) would select the cycles. Each of 26 channels (motor torque, torque sensor, joint position, and joint velocity on each of six joints, plus robot and system current) is resampled to 500 points over task progress and standardised against the 948 normal training cycles, $z = (x - \mu)/\max(\sigma, 0.05 \operatorname{range} \mu)$. For the retrieval, the record of a cycle keeps the five channels with the largest peak $|z|$, the dominant joint (the joint with the largest peak torque deviation in N m), and the action phase at that peak.

Knowledge base and retrieval. The knowledge base is an FMEA encoded as a graph. Liao et al. (2025) automate the construction of such graphs for robots; ours was written by the authors from the voraus-AD anomaly descriptions. It declares 14 failure modes. Twelve map one to one to the benchmark's categories, and two, backlash and overtemperature, have no instances. Each mode declares its joint scope, its signal families, a cycle-phase hint, the direction and time course of its torque deviation, and a recommended check. The score as built, which we call the deployed score, uses only the first three. A mode m is scored against a record by $s = J + F + 0.2E + 0.2P + 0.15C$. Here J is 1 if the dominant joint lies in the scope of m (0.5 for scope 'any'), and F is the Jaccard overlap of signal families. The term E is the mean over the record's five channels of $\log(14/n)\log(1 + |z|)$, where n is the number of modes that declare the channel's family and joint; a channel outside the families and joints of m contributes 0. Finally, P is 1 if the record's phase agrees with the phase hint of m, and C is the fraction of the families of m that E counts.

The form of the score and its weights were settled during development on these 755 cycles. To check whether this favours the retrieval, we chose the weights of E, P, and C again by MRR on each training half of the protocol below. The choice differs on every split, and on the test halves it gives top-1 0.176, MRR 0.362, and top-5 0.600, against 0.169, 0.366, and 0.651 for the deployed weights, all below the prior (Sec. 5.1).

Protocol. Unless stated otherwise, results are means over five grouped 50/50 splits, which we call leave-setting-out; Sec. 7 also uses leave-one-setting-out folds, each holding out a single setting. Each `setting` is assigned wholly to the training half or to the test half, stratified by category, so a learned component is always tested on instantiations it has not seen. Intervals are t-intervals over the five splits ($t_{0.975,4} = 2.776$). Because the splits re-partition the same 72 instantiations, the intervals describe the variation between splits on these data. Methods are compared split by split.

4. NON-ISOLABLE GROUPS

We first ask which modes the declarations can separate, before any cycle is scored. The fault signature matrix we analyse has one row per declaration that the score reads: joint scope, signal family, and phase. Fig. 1 shows the columns of the nine modes that share their joint scope and signal families with another mode. They form four groups: three load-torque modes (axis weight, lost can, varying weight), two distal collisions (foam, carton), two gripping faults (misgrip, invalid grip), and a commutation/overtemperature pair.

The form of the score decides whether the members of a group can still be told apart. Every term except phase agreement depends only on scope, families, and the record, so the members of a group can differ only in the phase term. For scope and families the score penalises a declaration broader than the evidence (extra families, scope 'any'), which corresponds to the exoneration assumption. For phase it does not: the hint 'any' receives full credit in both the grip phase (pick or place) and the transfer phase. In the collision, gripping, and commutation groups the phase hints are also equal, so the columns are identical and the members always receive the same score. In the load-torque group the hints differ (axis weight: any phase; lost can: transfer; varying weight: grip), so the column of

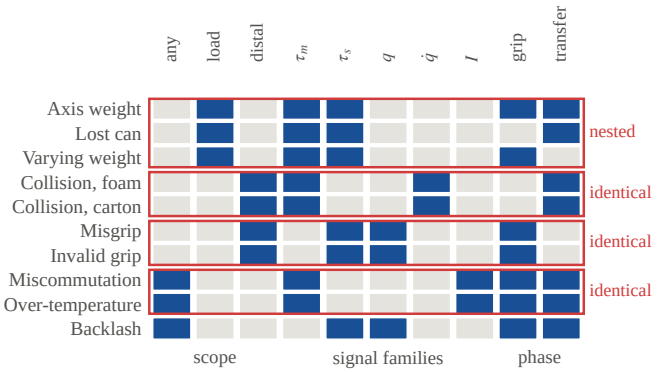


Fig. 1. Signature-matrix columns, drawn as rows, of the nine modes in non-isolable groups and of backlash, which has the gripping pair's families but not their scope. Scope 'load' is J1–J3, 'distal' J4–J6. Families: motor torque τ_m, torque sensor τ_s, joint position q, joint velocity $\dot{q}$, current I; a phase hint 'any' fills both phase cells.

axis weight contains the other two. Neither payload mode (lost can, varying weight) can then be isolated from axis weight. A record has one phase: in a grip phase axis weight ties with varying weight, in transfer with lost can, and in any other phase with both, so no member of the group ranks first alone.

The benchmark shows this pattern. In 514 of the 755 anomalies (68.1%) the top score is shared by two or three modes, and every tie lies inside one group. Of these ties, 498 are in the load-torque group: axis weight ties with lost can in 238 cycles and with varying weight in 199, and all three tie in 61 cycles whose peak lies outside the pick, place, and transfer phases. The other 16 are ties of the commutation pair. Setting the phase weight to any of seven values between 0 and 3 leaves the count at 514 and changes only which members tie. The collision and gripping groups never reach rank one. They are declared on the distal joints J4–J6, while the dominant joint, chosen by absolute torque deviation, is J1, J2, or J3 in every cycle.

These counts depend on how the knowledge base is written. If the phase of axis weight were declared as transfer instead of any, its 199 ties with varying weight would disappear and 315 ties would remain.

5. TWO REFERENCES FOR THE RANKED LIST

Because of the ties, top-1 accuracy depends on the tie-break order. We therefore give two references: a prevalence prior, and the range of top-1 over the 12 orders in which tied modes can be reported.

5.1 The prevalence prior

Against uniform chance, which is 0.071 over 14 modes, the retrieval has a top-1 lift of 2.37×. Uniform chance is a weak reference when categories differ in frequency, and on voraus-AD they differ widely: axis weight contributes 156 cycles and cable entanglement 10. The prevalence prior uses only these frequencies. It ranks the modes and groups by their frequency in the training half and gives the same ranking for every cycle.

Table 1. Retrieval and references on the five test halves. Group: accuracy of the predicted mode's group; for the prior, the most frequent group of the training half. Balanced: mean per-category recall, with tied modes sharing the credit; a constant scores $1/K$, with K the 10 or 11 categories in a test half. In parentheses: top-1 with tied modes sharing the credit. All orders: the retrieval's range over the 12 tie-break orders of Sec. 5.2. Two-stage: fixed {sign, step}, Sec. 6; flat classifier: Sec. 8.

	top-1	top-5	group	balanced
uniform chance	0.071	0.357	0.150	0.071
prevalence prior	0.187	0.705	0.401	0.095
retrieval	0.169 (0.160)	0.651	0.342	0.105
all orders	0.119–0.200	0.640–0.651	0.342	0.092–0.119
two-stage	0.309 (0.302)	—	0.342	0.208
flat classifier	0.525	—	0.574	0.397

Table 1 compares the prior with the retrieval. On top-3, top-5, MRR, and group accuracy the prior is ahead under every tie-break order of Sec. 5.2, and for top-3, MRR, and group accuracy it is ahead on all five splits. Top-1 is the exception, because its range over the orders contains the prior's value. Under the declared order the retrieval is 0.018 below the prior and behind on four of five splits, while two of the 12 orders put it at or above the prior. Breaking ties by training-half frequency, which uses only the information the prior uses, gives 0.200, above the prior's 0.187. For group accuracy, always naming the most frequent group of the training half, which is load torque on every split, gives 0.401 against 0.342 for the retrieval.

Balanced recall, the mean of the per-category recalls, is $1/K$ for a constant predictor, whatever the class frequencies. Over the test halves $1/K$ averages 0.095, so on this measure the prior has no advantage. The retrieval's balanced recall also depends on the tie-break, so we let tied modes share the credit: if the true mode is one of t modes tied at rank one, the cycle counts $1/t$. The retrieval then reaches 0.105, above the constant on every split by 0.002 to 0.016; under one of the 12 tie-break orders it falls below the constant. Its expected recall is zero on six of the 12 categories, four of them the collision and gripping modes that never reach rank one (Sec. 4), which keeps it close to the constant. Comparisons between methods below use balanced recall in this form.

5.2 The tie-break interval

The scores do not determine which member of a tied set is reported; our implementation reports the member declared first. Ties involve only five modes, always within their group, so the $3! \cdot 2! = 12$ relative orders of these modes cover every way the result can change. Table 1 gives the range over all of them: top-1 accuracy runs from 0.119 to 0.200, a width of 0.082. The declared order gives 0.169, and the mean over the orders, 0.160, equals the top-1 accuracy obtained when tied modes share the credit (McSherry and Najork, 2008).

This spread is larger than the variation due to the choice of split. Across the 12 orders, the five-split mean of top-1

has standard deviation 0.027 and range 0.082. Across the five splits, top-1 under the declared order has standard deviation 0.015 and range 0.040. More splits would not narrow the first spread, because one order applies to all splits. Both ranges are larger than the 0.018 by which the retrieval is below the prior, so top-1 alone does not settle which of the two is better. No tie-break rule can exceed 0.281, the top-1 of an oracle that always picks the correct member of the tied set. A rule that chooses inside the whole group can, because the phase term has already removed the true mode from 46 tied sets; the resolver of Sec. 6 is such a rule.

The group of the top-ranked mode does not depend on the order, because no tie crosses a group: group accuracy is 0.342 under every tie-break rule. Taken as a set, the group contains the true mode in 34.2% of cycles, with 2.34 modes on average. The same number of modes taken from the top of the prevalence list contains it in 38.5%. The group is ahead only on coverage averaged over categories (0.243 against 0.222). Reporting the group therefore gives an answer that does not depend on the tie-break order, but as a set it contains the true mode less often than the same number of modes chosen by frequency alone.

6. TEST SELECTION OVER REPRESENTATION DIMENSIONS

Sec. 4 showed that the deployed score cannot separate the members of a group. The knowledge base also declares the direction and time course of each mode's torque deviation, which the score does not use and which differ within every group except the collision pair. Adding them to the score with weight 0.2 removes 513 of the 514 ties, but it lowers top-1 from 0.169 to 0.148 on all five splits and balanced recall from 0.105 to 0.100 on four of them. Statistics of the existing signals must therefore be judged by their accuracy on held-out instantiations. For each non-isolable group we search for a repair, a set of such statistics. A classifier on these statistics, which we call the *resolver*, assigns a cycle of the group to one of its members.

Dimensions. The classifiers in this section read statistics of z. The retrieval uses only the peak magnitude $\max |z|$, and *mag* denotes this peak on each of the 26 channels. The candidate dimensions are further statistics of z: *sign* (mean of z), *step* (last third of the cycle minus first third), *payload* (carrying minus empty actions), *shape* (standard deviation of z), and *unread* (peak, signed, and payload statistics of two recorded channel families outside the 26). Each dimension has a cost class. It is *free* if it can be computed from the 26 channels, *pipeline* if it needs other recorded channels, and *hardware* if it needs a measurement the cell does not have. None of our dimensions is of hardware class.

Search and certificate. Every learned component is a standardised ℓ_2 logistic regression ($C = 1$). The search starts from mag and is greedy. At each step it adds the dimension whose gain in grouped inner cross-validation accuracy on the training half exceeds 0.005 times its cost factor (1 for free, 6 for pipeline) by the largest margin. It stops when no dimension has a gain above its threshold. The inner validation has $\min(4, n)$ folds, where n is the smallest number of training settings among the members. The selected set is adopted on a split if its inner accuracy exceeds the majority rate by 0.05 and the accuracy of mag alone by 0.02, where the majority rate is the accuracy of always naming the most frequent member of the training half. The repair is adopted for the group if this holds on a majority of splits. If a member has fewer than two instantiations in the training half, no search can run; we then evaluate the set of all free dimensions and do not adopt it. The test half is used only for the certificate, which gives the dimensions, their cost class, the held-out accuracy within the group, and three controls. Table 2 lists the accuracies. Varying the cost factor, the minimum gain, and both margins leaves the adoption decision unchanged for every group (results in the repository).

Adopted. Within the load-torque group the repair reaches 0.840 on held-out instantiations. All three controls are lower: the deployed features reach 0.562, the majority class 0.504, and the same dimensions with permuted labels 0.431. The repair is adopted on all five splits. The search selects sign on four of the five splits, step on three, and unread and shape on two each. With sign alone added to mag, the resolver reaches 0.827. Because unread enters on two splits, the worst-case cost class of the certificate is pipeline. The free set {sign, step}, however, reaches 0.869 against 0.872 for the best set with unread channels, so no additional channel is needed. With {sign, step}, most of the remaining errors are between the two payload modes. When the resolver has to choose between these two modes on their held-out cycles, it is right in 0.80 of cases, against 0.93 and 0.98 for the two pairs that include axis weight.

Not adopted. Carton collision has three instantiations, so each training half contains one of them and no grouped search can run. The set of all free dimensions reaches 0.776 against a majority rate of 0.718. Sec. 7 compares the protocols on this group with folds on which the search can run.

Undetermined. Each gripping mode has one instantiation, so no grouped split can test a repair.

Untestable. No over-temperature fault occurs in the benchmark, so no search is run. The recommended check for over-temperature names a statistic that would separate the pair: current against torque for the same motion.

The accuracies above are computed on cycles whose true mode is in the group. In use, the resolver sees only the cycles that the retrieval routes to the group, so we place it behind the retrieval. In this two-stage system, a cycle whose top group is load torque is passed to the resolver, and every other cycle keeps the retrieval's answer. With dimensions searched on each split, top-1 in declared order rises from 0.169 ± 0.018 to 0.302 ± 0.047. Fixing the dimensions selected on most splits, {sign, step}, gives 0.309 ± 0.031, a balanced recall of 0.208 (from 0.105), and 0.869 within the group. This choice was made with all five splits in view; choosing the dimensions from the other four splits each time gives 0.301 and 0.846, lower by 0.008 and 0.023. The gain of the fixed configuration over the retrieval is positive on every split, from 0.126 to 0.154.

The gain comes from the resolver, and the routing step costs accuracy. On the test halves the load-torque group is the true answer for 40.1% of cycles but is ranked first for 66.1%. The retrieval routes 69% of the load-torque

Table 2. Repair certificates: accuracy within the group on held-out cycles of its members. Grouped: leave-setting-out, the protocol we report. Ungrouped: the same code under a random split over cycles, including its inner validation. Deployed: the same classifier on the mag features alone. Majority: accuracy of always naming the training-half majority class. Permuted: the selected dimensions refitted on training labels shuffled over cycles. Deployed, Majority, and Permuted are grouped. †One carton instantiation in each training half, so no grouped search can run; the value is for the set of all free dimensions.

Group	Ungrouped	Grouped	Deployed	Majority	Permuted	Verdict
load torque (3 modes)	0.933	0.840	0.562	0.504	0.431	**adopted** on 5/5 splits, both protocols
distal collision (2 modes)	0.830	0.776†	0.709	0.718	0.608	**not adopted**: search cannot run; adopted ungrouped
gripping (2 modes)	one instantiation per member					undetermined
commutation/temp (2 modes)	no over-temperature instance in voraus-AD					untestable

cycles to the group, and 58% of the cycles it routes there are other faults. The resolver labels 37% of the routed cycles correctly, against 16% for the retrieval alone; on the load-torque cycles among them it is right in 88%. Sending every cycle to the load-torque resolver instead also reaches the 31% of load-torque cycles that the retrieval ranks elsewhere. It gives 0.349 and a balanced recall of 0.237, more than the two-stage system on all five splits. The collision and gripping groups are never ranked first (Sec. 4), so routing by top group would never apply a repair for them.

7. GROUPED AND RANDOM VALIDATION

The verdicts in Table 2 come from grouped splits. Because each `setting` value is one physical fault instantiation, a random split over cycles places cycles of the same instantiation in both halves, and a model can learn to recognise the instantiation instead of the fault. How much this matters depends on whether a method learns from labels. A flat 12-way logistic regression over all dimensions, the control of Sec. 8, falls from 0.800 under a random split to 0.525 under leave-setting-out, while the retrieval, which learns nothing from the training half, gives 0.161 and 0.162 over 20 splits of each kind.

The type of split also changes which repair is adopted. For the load-torque group the repair is adopted under both protocols (Table 2), but for the collision group the protocols disagree. Under a random split the inner validation, whose folds share instantiations, reaches 0.85 to 0.96, and the repair is adopted on all five splits. On held-out cycles, however, it exceeds the majority rate of the same split by only 3 of 47 cycles (0.830 against 0.766). Because no grouped search can run under the 50/50 leave-setting-out split (Sec. 6), and the group has nine instantiations, we compare the protocols with nine folds, the same search and adoption rule, and ten random seeds. With folds stratified over cycles, the repair is adopted on 9 of 9 folds, and its pooled held-out accuracy is 0.89 to 0.97. With one setting held out per fold (leave-one-setting-out), it is adopted on 4 of 9 folds, fewer than the five the rule requires, and its pooled held-out accuracy is 0.67 to 0.74, below the majority rate of 0.766 for every seed.

The permuted-label control of Table 2, which shuffles labels over cycles, does not detect this leakage. Under the random split it scores below the collision repair on all five splits (0.53 to 0.81 against 0.83), so the repair passes it. Labels shuffled over cycles are no longer tied to the instantiation, so the permuted model cannot gain from recognising instantiations, while the real model can. Permuting labels over whole settings keeps each label tied to an instantiation. Under the random split, 5 to 10 of the 84 possible labellings then reach the observed collision accuracy on each split ($p = 0.06$ to 0.12), so the setting-level control does not support the collision repair. For the load-torque repair, none of 300 labellings reaches the observed accuracy ($p < 0.004$). Comparisons of learned methods on this benchmark should therefore group by setting in the inner validation as well as in the outer split.

8. CONTROLS AND LIMITS

8.1 The flat classifier

The resolver uses features that the retrieval ignores, so we also compared the whole system with a flat classifier that reads these features without the knowledge base. Under the same protocol, a flat 12-way logistic regression over all dimensions reaches 0.525 top-1 against 0.309 for the two-stage system, and it is also ahead on balanced recall (0.397 against 0.208) and group accuracy (0.574 against 0.342). It recognises foam collision in 0.506 of cases, where the retrieval and the prior both score zero, so these features carry evidence that the retrieval does not use.

The knowledge base might still reduce the number of labels needed. We tested this over 20 paired leave-setting-out splits, training the classifier on k labelled cycles and the two-stage system on k labelled cycles per group. Only the load-torque resolver receives test cycles, so at most k labels affect the predictions of the two-stage system, as for the classifier. Both systems are scored on the 12-way task. At $k = 3$ the two-stage system, which is then close to the knowledge base alone, has the higher balanced recall on 14 of 20 splits. At $k = 8$ the classifier is ahead on 15 of 20 splits in both top-1 (0.225 against 0.208) and balanced recall (0.154 against 0.136). Its lead grows with k, and at $k = 320$ its top-1 is 0.550 against 0.304. If the two-stage system receives the same labelled cycles as the classifier, it is behind at $k = 8$ on all 20 splits. From eight labelled cycles on, the flat classifier is the more accurate of the two systems.

8.2 What would settle the open verdicts

Two verdicts rest on too few instantiations. Gripping would need at least two instantiations per mode for any grouped protocol, and voraus-AD has one. For collision, a 50/50 grouped split leaves $\lfloor N/2 \rfloor$ instantiations per mode for training, the grouped inner search needs two, and carton has $N = 3$, so the reported protocol needs a fourth carton instantiation. On the present data only leave-one-setting-out can run the search, and it does not adopt the repair (Sec. 7).

9. CONCLUSION

On voraus-AD, the ranked list of an FMEA-based diagnosis layer is limited by the structure of its knowledge base: 68.1% of anomalies tie at rank one, almost all in the load-torque group. A single top-1 figure then depends more on the tie-break order than on the split, and a prevalence prior is better on top-3, top-5, MRR, and group accuracy under every order.

Reporting the non-isolable group makes this limit visible. As an answer set, however, the group contains the true mode less often than the same number of modes chosen by frequency. A search over statistics of the existing signals then looks for ones that separate the group and reports them with held-out evidence, controls, and a cost class. For the load-torque group the selected statistics, most often the sign of the deviation, raise held-out accuracy within the group from 0.562 to 0.840, and the free set {sign, step} needs no additional channel. The search has to be validated by instantiation: under a random split it adopts a collision repair that leave-one-setting-out validation does not, and a permuted-label control over cycles passes it.

Isolability can be checked from the declarations before any data are seen, but the effect of removing ties on accuracy has to be measured. Scoring the declared direction and time course removes the ties and lowers accuracy, so a change of representation should be adopted on evidence from unseen instantiations. The same held-out evidence shows two limits of the knowledge base: routing by the retrieval is less accurate than sending every cycle to the resolver, and with eight or more labelled cycles a flat classifier over the same features is more accurate than the two-stage system. All results here come from one benchmark and one knowledge base written by the authors. The isolability check needs only the knowledge base and can run before deployment; the other checks need labelled cycles from several instantiations of each fault.

Code and data. Evidence records, certificates, all analyses, and the scripts that produce them are archived on Zenodo, doi: 10.5281/zenodo.22089173.

REFERENCES

Blanke, M., Kinnaert, M., Lunze, J., and Staroswiecki, M. (2016). *Diagnosis and Fault-Tolerant Control*. Springer, Berlin, 3rd edition.

Brockmann, J.T., Rudolph, M., Rosenhahn, B., and Wandt, B. (2024). The voraus-AD dataset for anomaly detection in robot applications. *IEEE Transactions on Robotics*, 40, 438–451.

Cordier, M.-O., Dague, P., Lévy, F., Montmain, J., Staroswiecki, M., and Travé-Massuyès, L. (2004). Conflicts versus analytical redundancy relations: a comparative analysis of the model based diagnosis approach from the artificial intelligence and automatic control perspectives. *IEEE Transactions on Systems, Man, and Cybernetics – Part B*, 34(5), 2163–2177.

de Kleer, J. and Williams, B.C. (1987). Diagnosing multiple faults. *Artificial Intelligence*, 32(1), 97–130.

Düstegör, D., Frisk, E., Cocquempot, V., Krysander, M., and Staroswiecki, M. (2006). Structural analysis of fault isolability in the DAMADICS benchmark. *Control Engineering Practice*, 14(6), 597–608.

Frisk, E., Krysander, M., and Åslund, J. (2009). Sensor placement for fault isolation in linear differential-algebraic systems. *Automatica*, 45(2), 364–371.

Hendriks, J., Dumond, P., and Knox, D. (2022). Towards better benchmarking using the CWRU bearing fault dataset. *Mechanical Systems and Signal Processing*, 169, 108732.

Jung, D., Frisk, E., and Krysander, M. (2015). Quantitative isolability analysis of different fault modes. *IFAC-PapersOnLine*, 48(21), 1275–1282.

Jung, D., Ng, K.Y., Frisk, E., and Krysander, M. (2018). Combining model-based diagnosis and data-driven anomaly classifiers for fault isolation. *Control Engineering Practice*, 80, 146–156.

Kapoor, S. and Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804.

Krysander, M., Åslund, J., and Nyberg, M. (2008). An efficient algorithm for finding minimal overconstrained subsystems for model-based diagnosis. *IEEE Transactions on Systems, Man, and Cybernetics – Part A*, 38(1), 197–206.

Krysander, M. and Frisk, E. (2008). Sensor placement for fault diagnosis. *IEEE Transactions on Systems, Man, and Cybernetics – Part A*, 38(6), 1398–1410.

Liao, X., Chen, C., Wang, Z., Liu, Y., Wang, T., and Cheng, L. (2025). Large language model assisted fine-grained knowledge graph construction for robotic fault diagnosis. *Advanced Engineering Informatics*, 65, 103134.

McSherry, F. and Najork, M. (2008). Computing information retrieval performance measures efficiently in the presence of tied scores. In *Advances in Information Retrieval (ECIR 2008)*, volume 4956 of *Lecture Notes in Computer Science*, 414–421. Springer, Berlin.

Reiter, R. (1987). A theory of diagnosis from first principles. *Artificial Intelligence*, 32(1), 57–95.

Travé-Massuyès, L. (2014). Bridging control and artificial intelligence theories for diagnosis: a survey. *Engineering Applications of Artificial Intelligence*, 27, 1–16.

Travé-Massuyès, L. and Wotawa, F. (2024). Bridging hardware and software diagnosis: leveraging fault signature matrix and spectrum-based fault localization similarities. In *35th International Conference on Principles of Diagnosis and Resilient Systems (DX 2024)*, volume 125 of *OASIcs*, 5:1–5:15.